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☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity

Agriculture/Animal Science

(8) Discipline/Field of Specialization
(9A) Abstract

This study was conducted at Agricultural Project and Research Area (APRA) at Brgy. CapSU, Dumarao, Capiz for three months. Generally, the research was conducted to determine the different dietary calcium sources in the performance, egg production and egg shell quality of quail. Specifically, it aimed to determine the egg production and egg shell quality in terms of the different calcium supplementation, determine which of the following calcium supplementation gives the best egg production and egg shell quality and lastly, to determine the income over feed cost of quail layers supplemented with oyster shell, chicken egg shell and their combination. The experimental birds were divided into 4 treatments with three replications each. Treatment 1 - control 100% commercial feeds, Treatment 2 were supplemented with 3% powdered oyster shell mixed in 97% commercial quail feed, Treatment 3 were supplemented with 3% powdered chicken egg shell mixed with 97% quail commercial feed and Treatment 4 were supplemented with 3% combined powdered oyster shell and chicken egg shell mixed with 97% quail commercial feeds. Completely Randomized design (CRD) was used in the study. Randomization was made using draw-lots scheme. Results of the study showed that the gain in weight, feed conversion ratio, and feed consumption were not affected by the different levels of ground egg shell and oyster shell. The mean egg weight, mean egg shell weight, mean shell thickness and mean egg production were significantly affected by the different levels of ground egg and oyster shell supplementation.

(9B) Keywords

Dietary calcium, Egg, Egg Shell, Oyster Shell, Quail,

(10) Introduction

Poor egg shell quality is a huge hidden cost to the egg producer. Estimates are that more than 10% of eggs produced in the hen house are uncollectible or break before intended use. The first 2-5 percent is lost simply, due to form which may be shell less, cracked or broken to the extent that they are not suitable for collection. Another 3-8 percent is lost during collection, moving through the belts, cleaning, packing and transportation to the end user. Because the first 2-5% loss is due to uncollectible eggs, most egg producers often estimate their egg loss due to poor shell quality at only this percentage, which is most likely an underestimation. At Rs.1.50/ egg, even a 5 percent loss could be as much as Rs. 2.7 million/year for 100 thousand layer house. The economic losses for the breeders will be even more due to reduced hatchability and chick liveability. Therefore, every effort must be directed towards improving shell quality and reducing egg breakage (<https://thepoultrysite.com/articles/maintaining-egg-shell-quality>). Date retrieved: 11/06/19

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Calcium (Ca) is the most important mineral for laying hens. Laying hens have high Ca requirements for bone maintenance and eggshell quality, which. Studies in previous years indicated that eggshell including membranes could be used as Ca source in layer diets as feed ingredient without adverse effects on productive performance, and could be an adequate Ca source in layer diets (Froning and Bergquist, 1990; Gongruttananun, 2011).

Although problems associated with shell quality have been studied extensively in the hen, very limited information is available in quails. As in the case of hen, a deficiency of calcium in the diet of quails cause decline in egg production. The optimum level of dietary calcium for high egg production and hatchability appears to be 2.5 to 3 percent, while higher level caused reduced hatchability (Sultana *et al.*, 2007).

There are lots of calcium sources available to choice as dietary supplement to boost up shell quality but still confrontation sometimes arises that which one to be most effective. With regard to the level of calcium, it is revealed that there is a gap of research with the intermediary level of 2.50 and 3. So the proposed level might be 2.5, 2.75 and 3(Sultana *et al.*, 2007).

Thus, there is a definite controversy regarding the level of calcium for quail. Therefore, more research is needed to solve this problem. Moreover, many works have been done only viewing shell quality. Very limited works have been undertaken regarding over all egg quality and egg production of laying quail.

Keeping all above facts in mind, the present study was undertaken for the purpose of identifying the comparative effects of oyster shell, chicken egg shell, and their combination in performance and egg quality of quail layers.

(11) Materials and Methods or Methodology

The materials used in the study were the following: sixty heads of quail layers, feeding and watering troughs, weighing scale, calculator and record book, oyster shell, egg shell, mortar and pestle, pale storage container.

Quality Assessment of quail egg

To determine the quality of **quail egg**, four treatments were utilized in the study and arranged in Completely Randomized Design (CRD) each treatment were replicated three times.

This treatment includes the following:

Treatment 1 - control 100% commercial feeds,

Treatment 2 were supplemented with 3% powdered oyster shell mixed in 97% commercial quail feed,

Treatment 3 were supplemented with 3% powdered chicken egg shell mixed with 97% quail commercial feed and

Treatment 4 were supplemented with 3% combined powdered oyster shell and chicken egg shell mixed with 97% quail commercial feeds.

Sixty (60)) day-old quail layers were procured from private quail raiser at Oton, Iloilo. Upon arrival of the birds, water with muscovado sugar was given to the birds to reduce transport stress.

a. To evaluate the quality of the quail egg the following parameters were used:

Egg Shell Quality

1. Egg weight.

2. Egg shell weight.

b. Layer performance

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Egg production

1. Initial weight.
2. Final weight.
3. Gain in weight.
4. Feed consumption
5. Feed conversion ratio.

Data collected were analyzed using the Analysis of Variance (ANOVA). LSD was used to determine the significant differences between treatment means at 5% level of significance.

(12) Results and Discussions

Table 1.0 presents the egg weight of quail layer on the 3rd week supplemented with different dietary calcium sources. Data shows that the birds fed with 3% powdered oyster shell have the highest egg weight with a mean of 10.57g. This was followed by the birds fed with 3% combined oyster and egg shell with a mean of 10.20g. This was followed by birds fed with 3% egg shell with a mean of 10.13g. Birds fed with pure commercial feeds attained the lowest egg weight with a mean of 9.90g.

Analysis of variance revealed that there was a significant difference among treatment means at 5% level of significance. LSD of the study reveals that quails fed with 3% oyster shell showed the highest results in egg weight.

Table 1.0. Mean Egg Weight (g) on the 3rd week

Treatment	Replication			Total	Mean
	1	2	3		
CF	10.00	9.80	9.89	29.69	9.90 ^a
3%OS + 97%CF	10.50	10.41	10.80	31.71	10.57 ^a
3%ES + 97% CF	9.89	10.35	10.15	30.39	10.13 ^b
3%OS&ES + 97%CF	10.26	10.45	9.89	30.60	10.20 ^{ab}
Grand Total				122.39	
Grand Mean					10.20

Table 2.0 presents the egg weight of quail layer on the 6th week supplemented with different dietary calcium sources. Data shows that the birds fed with 3% powdered oyster shell have the highest egg weight with a mean of 11.09g. This was followed by the birds fed with 3% egg shell with a mean of 11.01g. This was followed by birds fed with pure commercial feeds with a mean of 11.00g. Birds fed with combined oyster and egg shell attained the lowest egg weight with a mean of 10.97g.

Analysis of variance of the study revealed that there was no significant difference among treatment means. This means that the different levels of supplementation of dietary calcium sources did not affect the egg weight of the experimental birds.

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Table 2.0 Mean Egg Weight (g) on the 6th week

Treatment	Replication			Total	Mean
	1	2	3		
CF	11.08	10.74	11.17	32.99	11.00
3%OS + 97%CF	11.09	11.15	11.02	33.26	11.09
3%ES + 97% CF	11.12	11.01	10.91	33.04	11.01
3%OS&ES + 97%CF	11.09	10.69	11.14	32.92	10.97
Grand Total				132.21	
Grand Mean					11.02

Table 3.0 presents the egg weight of quail layer on the 9th week supplemented with different dietary calcium sources. Data shows that the birds fed with 3% powdered oyster shell have the highest egg weight with a mean of 11.18g. This was followed by the birds fed with 3% combined oyster and egg shell with a mean of 11.15g. This was followed by birds fed with 3% egg shell with a mean of 11.09g. Birds fed with pure commercial feeds attained the lowest egg weight with a mean of 10.94g.

Analysis of variance of the study revealed that there was no significant difference among treatment means. This means that the different levels of supplementation of dietary calcium sources did not affect the egg weight of the experimental birds.

Table 3. Mean Egg Weight (g) on the 9th week

Treatment	Replication			Total	Mean
	1	2	3		
CF	10.83	11.02	10.97	32.82	10.94
3%OS + 97%CF	11.12	11.24	11.18	33.54	11.18
3%ES + 97% CF	11.10	11.19	10.97	33.26	11.09
3%OS&ES + 97%CF	11.15	11.02	11.28	33.45	11.15
Grand Total				133.07	
Grand Mean					11.09

Table 4.0 presents the egg shell weight of quail layer on the 3rd week supplemented with different dietary calcium sources. Data shows that the birds under treatments 2, 3, and 4 have almost the

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same egg weight with a mean of 0.85g. This was followed by the birds fed with pure commercial feeds with a mean of 0.78g.

Analysis of variance of the study revealed that there was no significant difference among treatment means. This means that the different levels of supplementation of dietary calcium sources did not affect the egg shell weight of the experimental birds.

Table 4.0 Mean Egg Shell Weight (g) on the 3rd week

Treatment	Replication			Total	Mean
	1	2	3		
CF	0.81	0.78	0.76	2.35	0.78
3%OS + 97%CF	0.81	0.85	0.88	2.54	0.85
3%ES + 97%CF	0.82	0.89	0.84	2.55	0.85
3%OS&ES + 97%CF	0.88	0.83	0.85	2.56	0.85
Grand Total				10.00	
Grand Mean					0.83

Table 5.0 presents the egg shell weight of quail layer on the 6th week supplemented with different dietary calcium sources. Data shows that the birds fed with 3% powdered oyster shell have the highest egg shell weight with a mean of 0.90g. This was followed by the birds fed with 3% combined oyster and egg shell with a mean of 0.89g. This was followed by birds fed with 3% egg shell with a mean of 0.86g. Birds fed with pure commercial feeds attained the lowest egg shell weight with a mean of 0.83g.

Analysis of variance of the study revealed that there was no significant difference among treatment means. This means that the different levels of supplementation of dietary calcium sources did not affect the egg shell weight of the experimental birds.

Table 5.0 Mean Egg Shell Weight (g) on the 6th week

Treatment	Replication			Total	Mean
	1	2	3		
CF	0.83	0.81	0.86	2.50	0.83
3%OS + 97%CF	0.85	0.95	0.89	2.69	0.90
3%ES + 97%CF	0.86	0.88	0.83	2.57	0.86
3%OS&ES + 97%CF	0.87	0.91	0.88	2.66	0.89
Grand Total				10.42	
Grand Mean					0.87

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Table 6.0 presents the egg shell weight of quail layer on the 9th week supplemented with different dietary calcium sources. Data shows that the birds fed with 3% powdered oyster shell have the highest egg weight with a mean of 1.0g. This was followed by the birds fed with 3% combined oyster and egg shell with a mean of 0.99g. This was followed by birds fed with 3% egg shell with a mean of 0.92g. Birds fed with pure commercial feeds attained the lowest egg shell weight with a mean of 0.84g.

Analysis of variance of the study revealed that there is a highly significant difference among treatment means. LSD of the study reveals that quails fed with 3% oyster shell showed the highest results in egg shell weight.

Treatment	Replication			Total	Mean
	1	2	3		
CF	0.84	0.83	0.85	2.52	0.84 ^c
3%OS + 97%CF	0.96	1.00	1.03	2.99	1.00 ^a
3%ES + 97% CF	0.88	0.96	0.91	2.75	0.92 ^b
3%OS&ES + 97%CF	1.02	0.98	0.96	2.96	0.99 ^a
Grand Total				11.22	
Grand Mean					0.94

Table 7.0 presents the egg shell thickness of quail layer on the 3rd week supplemented with different dietary calcium sources. Data shows that the birds on treatment 2 and 4 have the highest egg shell thickness with a mean of 0.15mm. This was followed by the birds fed with 3% egg shell with a mean of 0.14mm. Birds fed with pure commercial feeds attained the lowest egg shell thickness with a mean of 0.11mm.

Analysis of variance of the study revealed that there was no significant difference among treatment means. This reveals that the calcium sources supplemented did not affect the egg shell thickness of the experimental birds.

Table 7.0 Mean Egg Shell Thickness (mm) on the 3rd week

Treatment	Replication			Total	Mean
	1	2	3		
CF	0.10	0.12	0.12	0.34	0.11
3%OS + 97%CF	0.15	0.15	0.14	0.44	0.15
3%ES + 97% CF	0.14	0.16	0.12	0.42	0.14
3%OS&ES + 97%CF	0.15	0.17	0.14	0.46	0.15
Grand Total				1.66	
Grand Mean					0.14

Table 8.0 presents the egg shell thickness of quail layer on the 6th week supplemented with different dietary calcium sources. Data shows that the birds on treatment 3 and 4 have the highest egg shell thickness with a mean of 0.19mm. This was followed by the birds fed with 3% oyster

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shell with a mean of 0.17mm. Birds fed with pure commercial feeds attained the lowest egg shell thickness with a mean of 0.11mm.

Analysis of variance of the study revealed that there is a significant difference among treatment means at 1% level of significance. LSD of the study reveals that quails fed under treatments 3 and 4 showed the highest results in egg shell thickness.

Table 8. Mean Egg Shell Thickness (mm) on the 6th week

Treatment	Replication			Total	Mean
	1	2	3		
CF	0.12	0.12	0.10	0.34	0.11 ^a
3%OS + 97%CF	0.18	0.18	0.15	0.51	0.17 ^a
3%ES + 97% CF	0.18	0.18	0.20	0.56	0.19 ^a
3%OS&ES + 97%CF	0.16	0.20	0.20	0.56	0.19 ^a
Grand Total				1.97	
Grand Mean					0.16

Table 9.0 presents the egg shell thickness of quail layer on the 9th week supplemented with different dietary calcium sources. Data shows that the birds on treatment 2 and 4 have the highest egg shell thickness with a mean of 0.21mm. This was followed by the birds fed with 3% egg shell with a mean of 0.19mm. Birds fed with pure commercial feeds attained the lowest egg shell thickness with a mean of 0.12mm.

Analysis of variance of the study revealed that there was no significant difference among treatment means at 5% level of significance. LSD of the study reveals that quails fed under treatments 2 and 4 showed the highest results in egg shell thickness.

Table 9.0 Mean Egg Shell Thickness (mm) on the 9th week

Treatment	Replication			Total	Mean
	1	2	3		
CF	0.13	0.10	0.13	0.36	0.12 ^a
3%OS + 97%CF	0.25	0.20	0.18	0.63	0.21 ^a
3%ES + 97% CF	0.18	0.20	0.18	0.56	0.19 ^a
3%OS&ES + 97%CF	0.25	0.18	0.20	0.63	0.21 ^a
Grand Total				2.18	
Grand Mean					0.18

Table 10.0 presents the egg production of quail layer on the 3th week supplemented with different dietary calcium sources. Data shows that the birds fed with 3% combined oyster and egg shell have the highest egg production with a mean of 93. This was followed by the birds fed with pure commercial feeds with a mean of 92. This was followed by the birds fed with 3% oyster shell with a mean of 91. Birds fed with 3% egg shell attained the lowest egg shell thickness with a mean of 88.

Analysis of variance of the study revealed that there is a significant difference among treatment means at 5% level of significance. LSD of the study reveals that quails fed with 3% combined oyster and egg shell showed the highest results in egg production.

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Table 10.0 Mean Egg Production on 3rd week

Treatment	Replication			Total	Mean
	1	2	3		
CF	93	92	90	275	92 ^a
3%OS + 97%CF	90	90	93	273	91 ^a
3%ES + 97%CF	87	89	89	265	88 ^a
3%OS&ES + 97%CF	92	92	94	278	93 ^a
Grand Total				1091	
Grand Mean					91

Table 11.0 presents the egg production of quail layer on the 6th week supplemented with different dietary calcium sources. Data shows that the birds fed with 3% combined oyster and egg shell have the highest egg production with a mean of 96. This was followed by the birds under treatment 2 and 3 with the mean of 96. Birds fed pure commercial feeds attained the lowest egg shell thickness with a mean of 88.

Analysis of variance of the study revealed that there is a significant difference among treatment means at 1% level of significance. LSD of the study reveals that quails fed with 3% combined oyster and egg shell showed the highest results in egg production.

Table 11.0 Mean Egg Production on 6th week

Treatment	Replication			Total	Mean
	1	2	3		
CF	89	91	92	272	91 ^a
3%OS + 97%CF	97	95	97	289	96 ^a
3%ES + 97%CF	96	94	97	287	96 ^a
3%OS&ES + 97%CF	99	98	94	291	97 ^a
Grand Total				1139	
Grand Mean					95

Table 12.0 presents the egg production of quail layer on the 9th week supplemented with different dietary calcium sources. Data shows that the birds fed with 3% oyster shell have the highest egg production with a mean of 98. This was followed by the birds fed with 3% combined oyster and egg shell with the mean of 95. This was then followed by the birds fed with 3% egg shell with the mean of 94. Birds fed with pure commercial feeds attained the lowest egg shell thickness with a mean of 92.

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Analysis of variance of the study revealed that there is a significant difference among treatment means at 5% level of significance. LSD of the study reveals that quails fed with 3% oyster shell showed the highest results in egg production.

Table 12.0 Mean Egg Production on 9th week

Treatment	Replication			Total	Mean
	1	2	3		
CF	93	88	94	275	92 ^a
3%OS + 97%CF	98	99	98	295	98 ^a
3%ES + 97%CF	92	93	96	281	94 ^a
3%OS&ES + 97%CF	97	95	94	286	95 ^{ab}
Grand Total				1137	
Grand Mean					95

Table 13.0 presents the gain in weight of quail layer supplemented with different dietary calcium sources. Data shows that the birds fed with 3% egg shell have the highest gain in weight with a mean of 26.61g. This was followed by the birds fed with 3% oyster and egg shell with the mean of 24.51. and by the birds fed with 3% oyster shell with the mean of 22.51g, respectively. Birds fed with pure commercial feeds obtained the lowest gain in weight with a mean of 17.54g.

Analysis of variance revealed that there is a significant difference among treatments. This means that different levels of supplementation of dietary calcium sources affects the gain in weight of experimental birds.

Table 13.0 Mean gain in weight (g)

Treatment	Replication			Total	Mean
	1	2	3		
CF	22.48	12.68	17.46	52.62	17.54
3%OS + 97%CF	20.00	25.36	21.10	66.46	22.15
3%ES + 97%CF	23.06	30.68	26.08	79.82	26.61
3%OS&ES + 97%CF	24.12	29.12	20.30	73.54	24.51
Grand Total				272.44	
Grand Mean					22.70

The mean feed consumption of quail layer fed with different dietary calcium sources in Table 14.0 reveals that the highest feed consumption was observed from the birds that were given 100% with a mean of 27.00g. This was followed by birds given with 3% combined oyster and egg shell with a mean of 26.67g, while the lowest feed consumption was observed from the birds fed with 3% ES with a mean of 26.64g.

The analysis of variance shows that the supplementation of different dietary calcium supplements do not significantly affect the mean feed consumption of quail layer at 1% level of significance.

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Table 14.0 Mean Feed Consumption (g)

Treatment	Replication			Total	Mean
	1	2	3		
CF	27.02	27.06	26.93	81.01	27.00 ^a
3%OS + 97%CF	26.61	26.69	26.72	80.02	26.67 ^a
3%ES + 97%CF	26.64	26.64	26.65	79.93	26.64 ^a
3%OS&ES + 97%CF	26.68	26.64	26.64	79.96	26.65 ^a
Grand Total				320.92	
Grand Mean					26.74

The mean feed conversion ratio of quail layer fed with different dietary calcium sources in Table 14.0 reveals that the best conversion ratio was observed from the birds that were given 3% egg shell with a mean of 1.02g. This was followed by birds given with 3% combined oyster and egg shell with a mean of 1.13g, while the poorest feed conversion ratio was observed from the birds fed with pure commercial feeds with a mean of 1.51g.

The analysis of variance shows that the supplementation of different dietary calcium supplements did not significantly affect the mean feed conversion ratio of quail layer.

Table 15.0 FCR

Treatment	Replication			Total	Mean
	1	2	3		
CF	1.20	2.13	1.20	4.53	1.51
3%OS + 97%CF	1.33	1.05	1.27	3.65	1.22
3%ES + 97%CF	1.16	0.87	1.02	3.05	1.02
3%OS&ES + 97%CF	1.16	0.92	1.31	3.39	1.13
Grand Total				14.62	
Grand Mean					1.22

The net income of quail layer as affected with different dietary calcium supplements presented in table 16.0 Data showed that birds given with 3% oyster shell had the highest net income of Php 1146.85, followed by the birds given 3% combined oyster and egg shell with a net income of Php 1142.85 and birds given 3% egg shell with a net income of Php 1100.85. Lowest net income was noted on birds given pure commercial feeds with a net income of Php 1076.85.

From the result, it can be recommended to use mix 3% oyster shell in their ration since the layer can have higher net income compared to layers fed with pure commercial feed.

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Table 16.0 Income over Feed Cost of Quail Layers fed with Different Dietary Calcium Sources

PARTICULAR	TREATMENT			
	CONTROL	3%OS + 97%CF	3%ES + 97%CF	1.5%OS + 1.5%ES + 97%CF
Sales of eggs and birds	2544	2614	2568	2610
Production cost				
Cost of birds	600	600	600	600
Cost of feeds	822.15	822.15	822.15	822.15
Cost of water	45	45	45	45
Sub total	1467.15	1467.15	1467.15	1467.15
Net income	1076.85	1146.85	1100.85	1142.85

(13) Conclusions and Recommendations

Based on the results of the study, the following conclusions were made:

1. Oyster and egg shell are effective feed supplement to the egg production and egg shell quality of the quail layer.
2. Oyster shell and egg shell are both effective feed supplements in terms of egg weight, shell thickness, egg production and feed consumption.
3. All of the treatments are profitable.

Recommendations

Based on the result of the study, the following recommendations are made:

1. Use oyster shell and egg shell as feed supplement to quail layer to increase the egg shell quality and egg production.
2. Further studies on feeding trials of the oyster and egg shell in other forms of preparation may be conducted.

(14) Output

Product	Quail egg
Publication	IEC materials
Patent	Utility models for dietary calcium levels in the egg production of quail
People and Services	Helping farmers with their quail farming business
Places and Linkages	Partnership with the quail farmers, LGU and oyster producers. Linkages with the DTI and LGU for the marketing of the product.

(15) Technology Transfer Activity Conducted

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(16) Economic Analysis

(17) Acknowledgments

(19) Literature Cited

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